25 August 1976
SUPERSEDING
MIL-C-23292A
10 September 1968

MILITARY SPECIFICATION

CONTROLS, TEMPERATURE, REFRIGERANT-12, WITH HIGH PRESSURE CUT-OUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers temperature controls for low temperature control and high pressure safety cut-out of refrigerant-12 compressors.

1.2 Classification. Temperature controls covered by this specification shall be of the following types and classes as specified (see 6.2).

Type I	-	120/240-volt, 50 to 60 Hertz alternating-current rating.
Type II	-	24-volt, direct-current rating.
Class 1	-	Automatic reset for high pressure safety cut-out.
Class 2	-	Manual reset for high pressure safety cut-out.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4130/5930

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PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Set-Up.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291 - Paperboard, Wrapping and Cushioning.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS 17844 - Controls, Temperature, Refrigerant-12; with High Pressure Cut-Out.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 873 - Temperature-Indicating-and-Regulating Equipment
Electrical Appliance and Utilization Equipment List

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Boulevard, Tampa, FL 33619.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Temperature controls delivered under this specification shall be new and the manufacturer's current commercial product, except for any changes necessary to comply with specification requirements.

3.2 Standard. Controls shall conform to the applicable requirements of UL 873.

3.2.1 Certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the controls he proposes to supply under this specification meet the applicable requirements of UL 873. Acceptable evidence of meeting the requirements of the UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the control offered has been tested and conforms to the applicable requirements of UL 873.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial temperature control, with high pressure cut-out.

3.4 Design and construction. The temperature control shall be of the types and classes listed in 1.2, and as specified in MS 17844. The type I control shall operate on a 120/240 volt, 50 to 60 Hertz alternating current. The type II control shall operate on a 24 volt, direct-current rating. The class 1 shall have an automatic reset for high pressure safety cut-out. The class 2 shall have a manual reset for high pressure safety cut-out. The temperature control shall open and close to stop and start the compressor between temperature limits of minus 10 degrees Fahrenheit (F.) (-8 celsius (c)) to 55 F. (13 C.) or 50 F. (10 C.) to 90 F. (32 C.), as applicable, in accordance with the control setting (see 4.3.1).

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* 3.5 Marking for identification. The temperature control shall be permanently marked with the manufacturer's name, trade name, or trademark in such known characters as to be identifiable with said manufacturer and in accordance with UL 873.

3.6 Finish. The temperature control shall have the manufacturers' standard finish.

3.7 Workmanship. The finished temperature control shall be free from defects; marred surfaces, dents, punctures, fractures, rough edges, burrs, slivers. All welds or soldering shall be smooth and uniform, completely fused, without undercuts, cracks, burnholes, or fractures. Fastener threads shall not be broken or stripped. All fasteners shall be tight. No component shall be missing, or malformed. All components shall be carefully assembled, adjusted and aligned.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall be the number of temperature controls of one type and class offered for acceptance at one time. The sample unit for this inspection shall be one completely assembled temperature control.

4.2.2.1 Visual examination. Examination of the end item shall be made for defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's standard finish.		X
Design, construction and workmanship	Fasteners, loose, broken or with stripped threads.	X	
	Part missing or not as specified.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel.	X	
	Any component fractured, split, punctured, dented, or otherwise damaged.	X	
Welding or soldering (when applicable)	Missing, not adherent, or incomplete.	X	
	Cracks, burnholes or fractures, not smooth, uniform, completely fused.	X	
Assembly	Not properly assembled or secured.	X	
Marking	Missing, incomplete, not legible.	X	

4.2.2.2 Dimensional examination. Dimensional examination shall be made of the end item to determine compliance with this specification. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. Each production item shall be tested in accordance with 4.3.1. Failure to pass the tests shall be cause for rejection of the item.

4.2.3 Standards compliance. Proof of compliance with UL requirements of 3.2 shall be available to the Government representative.

* 4.2.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and markings are in compliance with section 5 of this specification. Defects shall be scored as specified below. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects expressed in terms of defects per hundred units.

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* TABLE II. Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or not as specified.
Content	Number of packages per shipping container is more or less than required.

4.2.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.3 Tests.

* 4.3.1 Operation test. The control shall be subjected to conditions encountered in actual use to determine if contacts open and close according to the various settings specified on MS 17844 (see 3.4). Tests shall be performed for both the temperature control and the high pressure safety cut-out feature of the control in accordance with the manufacturer's standard practice. Failure of the control to operate on temperature or high pressure, or failure of the contacts to open or close shall constitute failure of this test.

* 5. PACKAGING

* 5.1 Preservation-packaging. Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Each control shall be packaged in a snug-fitting box conforming to variety 1, style II, III or VII of PPP-B-566; style optional of PPP-B-665; type I, variety 1, class A, style 2 of PPP-B-676; or style optional, type CF, variety SW, grade V3c or type SF, class domestic of PPP-B-636. Prior to boxing each control shall be completely wrapped in cellulosic cushioning material conforming to type II, class B of PPP-C-843; paperboard wrap conforming to type I, style 1 or 2 of PPP-P-291, or provided with a snug-fitting sleeve fabricated from fiberboard conforming to type CF, variety SW, class domestic of PPP-F-320, in such a manner as to prevent damage while in transit. Box closure shall be in accordance with the applicable box specification.

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5.1.2 Level C. Each control shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Six controls, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type SF, grade V2s, class weather-resistant, of PPP-B-636. Level A packages shall be packed on end, three in length, two in width, and one in depth. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Six controls, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Level A packages shall be packed on end, three in length, two in width, and one in depth. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 Weather-resistant boxes. When specified (see 6.2) the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C. Controls, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The temperature controls, covered by this specification are for automatic, cyclic ("ON" and "OFF") of motor-driven refrigerant-12, compressors. The safety cut-out provides protection against excessively high refrigerant pressures.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(d) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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